

Density-Dependent Coherence Integration: Detailed Derivation of Conscious Coherence Term in Effective Density - v13.0 FINAL

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Abstract

This paper presents the corrected derivation of the conscious coherence contribution to effective energy density $\rho_{\text{eff}}(a)$ in Horndeski-based EFT of Dark Energy ($G_5=0$, $G_4=G_4(\phi)$). Previous versions v1-v11 coupled ρ_{psi} via multiplicative f_L . v13 corrects to additive coupling $\rho_{\text{psi}} = \xi \epsilon_0 \Phi_t$ [kg/m³], $\xi \epsilon_0 = 1.0e-26$ kg/m³ reference, $\Phi_t = 0.6 \cdot \text{EEG_coh} + 0.4 \cdot \text{HRV_coh}$ in [0,1]. Background corrected: $\rho_m(a) = \rho_{m0} a^{-3}$ replaces ρ_L . $r(a) = G_4 \phi / (X G_3 X - G_4 \phi) = 1/(1+\epsilon(a))$, $\epsilon(a) = X G_3 X / G_4 \phi - 2$, $R(a) = (1+r(a))/(1+2r(a))$, $dR/dr = -1/(1+2r)^2 < 0$. Model is separately falsifiable by Euclid/DESI ($h_{\text{class}} 0.03\%$ Planck) and EEG/HRV correlation.

1. Introduction

Time Attraction v1-v11 introduced $f_L = \exp(-\gamma_L \rho_L)$, $\gamma_L = G_N L^2/c^2$, lab prediction $\Delta f/f > 1e-18$ for $\Delta \rho = 1e4$ kg/m³. Previous coupling $\rho_{\text{psi}} * f_L(R)$ used $R_{\text{old}} = (r+1)/(r+2)$, causing 25-36% systematic error. This work implements corrected $R(a) = (1+r(a))/(1+2r(a))$, decouples f_L from background EFT functions $\alpha_M(a)$, $\alpha_B(a)$, with full denominator in α_B .

2. Coherence Metrics and Φ_t

EEG_coh in [0,1]: alpha band 8-12 Hz inter-hemispheric coherence $|S_{xy}|/\sqrt{S_{xx}S_{yy}}$ or PLV, 5 min, 19 channels. 0=incoherent, 1=fully coherent. HRV_coh in [0,1]: heart coherence ratio peak power/total 0.04-0.26 Hz or normalized RMSSD, 5 min EKG/PPG. 0=chaotic, 1=sinusoidal 0.1 Hz. Weights from cortical dominance 60/40, normalized $w_{\text{EEG}} + w_{\text{HRV}} = 1$.

$$\Phi_t = 0.6 * \text{EEG_coh} + 0.4 * \text{HRV_coh} \text{ in } [0,1] \quad (1)$$

3. Physical Density ρ_{psi}

ϵ_0 is normalization, ξ coupling scale. Reference $\xi \epsilon_0 = 1.0e-26$ kg/m³ sets order of magnitude. Example: EEG_coh=0.70, HRV_coh=0.80 => $\Phi_t=0.74$ => $\rho_{\text{psi}}=7.4e-27$ kg/m³.

$$\rho_{\text{psi}} = \xi * \epsilon_0 * \Phi_t \text{ [kg/m}^3\text{]} \quad (2)$$

4. Background Correction and $R(r)$ - Own Derivation

Correction: ρ_L replaced by $\rho_m(a) = \rho_{m0} a^{-3}$, $a = 1/(1+z)$. $r(a)$ independent of f_L : $r = \alpha_0 f_L / [\alpha_0 f_L (1+\epsilon)] = 1/(1+\epsilon)$.

$$M^2 = 2 G_4$$

$$\alpha_M = 2 \phi_{\text{dot}} G_4 \phi / (H M^2)$$

$$\alpha_B = 2 \phi_{\text{dot}} [X G_3 X - G_4 \phi] / (H M^2) \text{ (corrected, full denominator)}$$

$$R(a) = (1+r(a))/(1+2r(a)), \quad dR/dr = -1/(1+2r)^2 < 0 \quad (3)$$

$$r(a) = (1-R)/(2R-1) = G_4 \phi / (X G_3 X - G_4 \phi) = 1/(1+\epsilon(a)) \quad (4)$$

$$\epsilon(a) = X G_3 X / G_4 \phi - 2 \quad (5)$$

$$\text{Locus: } \epsilon=0 \Leftrightarrow X G_3 X = 2 G_4 \phi \Leftrightarrow r=1 \Leftrightarrow R=2/3 \quad (6)$$

5. Integration into $\rho_{\text{eff}}(a)$

Previous error: multiplicative $\rho_{\text{psi}} * f_L$ invalid because f_L not background EFT function. Corrected additive form: All $M_i(R(a))$ depend on background $R(a)$ only. $\Phi_t(a)$ is binned temporally and correlated with $a(t)$ without multiplicative coupling. This restores EFT consistency.

$$\rho_{\text{eff}}(a) = \rho_{b0} M_{\text{Rho}}(R(a)) + \rho_{DM0} \rho(R(a)) + \rho_{B0} M_B/R(a) + \rho_{GR} M_B/R(a) + \rho_{E0} M_B/R(a) + \rho_{\text{psi}}(\Phi_t) \quad (7)$$

6. Full Formula Collection v13.0

$R = (1+r)/(1+2r)$

$r = G4_phi/(X\ G3X - G4_phi) = 1/(1+epsilon)$

$epsilon = X\ G3X/G4_phi - 2$

$M^2 = 2\ G4, \alpha_M = 2\ phi_dot\ G4_phi/(H\ M^2)$

$\alpha_B = 2\ phi_dot\ [X\ G3X - G4_phi]/(H\ M^2)$

$\rho_m(a)=\rho_{m0}\ a^{-3},\ a=1/(1+z)$

$\Phi_t = 0.6*EEG_coh + 0.4*HRV_coh$

$\rho_{psi} = \Xi*\epsilonpsilon_0*\Phi_t$

$\rho_{eff} = \sum \rho_0*M_i(R) + \rho_{psi}$

$f_L = \exp(-\gamma_L\ \rho_m(a)),\ \gamma_L=G_N\ L^2/c^2$

Mennyiség	Kifejezés	Példa
R	$(1+r)/(1+2r)$	$r=0.5 \rightarrow 0.75$
r	$1/(1+eps)$	$eps=0 \rightarrow 1$
Phi_t	$0.6*EEG+0.4*HRV$	0.74
rho_psi	$\Xi*\epsilonps_0*\Phi_t$	$7.4e-27\ kg/m3$
rho_eff	$\sum M_i(R)+\rho_{psi}$	additív

7. Falsifiability and Conclusion

Cosmology: model fails if $R(a) \neq (1+r)/(1+2r)$ or hi_class not 0.03% Planck recovery with $r=1..1.3$ in Euclid/DESI 4x3 tomographic bins. Consciousness: fails if Φ_t shows no correlation with $a(t)$ or $\rho_{psi}=0$, i.e. EEG_coh and HRV_coh random. Laboratory: $f_L = \exp(-\gamma_L\ \rho_m(a))$, $\Delta \rho=1e4\ kg/m3$ must show $\Delta f/f > 1e-18$ (lead-vacuum 1.44 ns/s, tungsten-vacuum 2.45 ns/s) with atomic clocks. Conclusion: v13.0 provides corrected, EFT-consistent, additive coupling $\rho_{eff}(a)= \sum \rho_0\ M_i(R(a)) + \rho_{psi}(\Phi_t)$, with $R=(1+r)/(1+2r)$, $r=1/(1+epsilon)$, $epsilon=X\ G3X/G4_phi - 2$, $\rho_m(a)$ background, and separately testable predictions.

References

[1] Bellini & Sawicki 2014 Eq.2.18 (for comparison, not used in own derivation)

[2] Planck 2018

[3] hi_class , Euclid, DESI

[4] EEG coherence $|S_{xy}|/\sqrt{S_{xx}*S_{yy}}$, HRV coherence 0.04-0.26 Hz

[5] Own derivation $G5=0, G4=G4(phi)$